

Assignment - 4

1. Write a C program to check whether a year is leap year or not.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int year;
```

```
    printf("Enter a year");
```

```
    scanf("%d", &year);
```

```
    if ((year % 400 == 0) || ((year % 4 == 0) && (year % 100 != 0)))
```

```
    {
```

```
        printf("Leap year");
```

```
    }
```

```
    else { printf("Not a leap year"); }
```

```
    return 0; }
```

Output

Enter a year 2012

Leap year

2. Write a C program to check whether a character is uppercase or lowercase.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char c;
```

```
    printf("Enter the character");
```

```
    scanf("%c", &c);
```

```
    if (c >= 65 && c <= 90)
```

```
    {
```

```
        printf("Uppercase");
```

```
    }
```

```
    else {
```

```
        printf("Lowercase");
```

```
    }
```

```
    return 0;
```

```
}
```

Output

Enter the character A

Uppercase.

3. Write a C program to input week number and print week day.

```
#include <stdio.h>
```

```
{ int main( )
```

```
{ int num;
```

```
printf ( "Enter a number " );
```

```
scanf ( " %d", &num );
```

```
switch ( num ) {
```

```
case 1:
```

```
printf ( "Sunday" );
```

```
break;
```

```
case 2:
```

```
printf ( "Monday" );
```

```
break;
```

```
case 3:
```

```
printf ( "Tuesday" );
```

```
break;
```

```
case 4:
```

```
printf ( "Wednesday" );
```

```
break;
```

```
case 5:
```

```
printf ( "Thursday" );
```

```
break;
```

```
case 6:
```

```
printf ( "Friday" );
```

```
break;
```

```
case 7:
```

```
printf ( "Saturday" );
```

```
break;
```

```
default:
```

```
printf ( "Invalid number" );
```

```
}
```

```
return 0;
```

```
}
```

Output

Enter a number 1

Sunday

4. Check whether alphabet is vowel or consonant using switch case.

```
#include <stdio.h>
int main()
{
    char ch;
    printf("Enter a character");
    scanf("%c", &ch);
    switch(ch) {
        case 'A': case 'a': case 'E': case 'e': case 'I': case 'i':
        case 'O': case 'o': case 'U': case 'u':
            printf("Vowel");
            break;
        default:
            printf("Consonant");
    }
    return 0;
}
```

Output

Enter a character A
Vowel

5. Write a C program to input month number and print number of days in that month.

```
#include <stdio.h>
int main()
{
    int num;
    printf("Enter the number");
    scanf("%d", &num);
    switch(num) {
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            printf("Month has 31 days");
            break;
        case 2:
            printf("Month has 28 days"); break;
        case 4: case 6: case 9: case 11:
            printf("Month has 30 days");
            break;
        default:
            printf("Invalid number");
    }
    return 0;
}
```

Output → Enter the number 10 | Month has 31 days

8. Write a program to count total numbers of notes in given amount.

```
#include <stdio.h>
```

```
int main()
{
    int n, count;
    printf("Enter the amount ");
    scanf("%d", &n);
    count = n / 500;
    n = n % 500;
    printf("500 = %d \n", count);
    count = n / 200;
    n = n % 200;
    printf("200 = %d \n", count);
    count = n / 100;
    n = n % 100;
    printf("100 = %d", count);
    count = n / 50;
    n = n % 50;
    printf("50 = %d", count);
    count = n / 20;
    n = n % 20;
    printf("20 = %d", count);
    count = n / 10;
    n = n % 10;
    printf("10 = %d", count);
    return 0;
}
```

Output

Enter the amount 1300

500 = 2

200 = 1

100 = 1

50 = 0

20 = 0

10 = 0

7. Write a C program to input marks of five subject physics, Chemistry, Biology, Mathematics and computer. Calculate percentage and grade according to following:

percentage $\geq 90\%$. A+

percentage $\geq 80\%$. A

percentage $\geq 70\%$. B

percentage $\geq 60\%$. C

percentage $\geq 40\%$. D

percentage $< 40\%$. E

Grade F

```
#include <stdio.h>
```

```
{ int main()
```

```
{ int sub1, sub2, sub3, sub4, sub5, total;  
float per;
```

```
printf("Enter the five subject mark");
```

```
scanf("%d %d %d %d %d", &sub1, &sub2, &sub3, &sub4, &sub5);
```

```
total = sub1 + sub2 + sub3 + sub4 + sub5;
```

```
per = total * 5.0;
```

```
if (per  $\geq 90$ ) { printf("Grade A+"); }
```

```
else if (per  $\geq 80$ ) { printf("Grade A"); }
```

```
else if (per  $\geq 70$ ) { printf("Grade B"); }
```

```
else if (per  $\geq 60$ ) { printf("Grade C"); }
```

```
else if (per  $\geq 40$ ) { printf("Grade D"); }
```

```
else if (per  $< 40$ ) { printf("Grade E"); }
```

```
else { printf("Grade F"); }
```

```
return 0;
```

```
}
```

Output

Enter five subject mark

50 70 60 80 90

Grade B

8. Write a c program to input basic salary of an employee and calculate its gross salary according to following: Basic salary ≤ 10000 : HRA = 20%, DA = 80%.

Basic salary ≤ 20000 : HRA = 25%, DA = 90%.

Basic salary > 20000 : HRA = 30%, DA = 95%.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
float basic, hra, da, gross;
```

```
printf("Enter basic salary ");
```

```
scanf("%f", &basic);
```

```
if (basic  $\leq$  10000) {
```

```
    hra = basic * 0.20;
```

```
    da = basic * 0.80;
```

```
}
```

```
elseif (basic  $\leq$  20000) {
```

```
    hra = basic * 0.25;
```

```
    da = basic * 0.90;
```

```
}
```

```
else {
```

```
    hra = basic * 0.30;
```

```
    da = basic * 0.95;
```

```
}
```

```
gross = basic + hra + da;
```

```
printf("Basic salary = %f", basic);
```

```
printf("HRA = %f", hra);
```

```
printf("DA = %f", da);
```

```
printf("Gross salary = %f", gross);
```

```
return 0;
```

```
}
```

Output

Enter basic salary 24000

Basic salary = 24000.00

HRA = 7200.00

DA = 22800.00

Gross salary = 54000.00

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